mergencystretch

Chapter 6. Potential Outcomes and Effect Estimation

Potential outcomes, IPW and doubly robust estimation, CATE meta-learners, the R-learner

Contents

6.1 Potential outcomes **2**
6.1.1 Frameworks for causal inference . . . 2
6.1.2 Basic concepts . . . 2
6.1.2.1 The fundamental problem of causal inference . . . 2
6.1.2.2 God's eye view: individual versus average effects . . . 3
6.1.2.3 What the researcher actually sees . . . 3
6.1.3 Assumptions linking potential and observed outcomes . . . 3
6.1.3.1 SUTVA . . . 3
6.1.3.2 Consistency and well-defined interventions . . . 4
6.1.4 Causal estimands . . . 4
6.1.5 Randomization and observational studies . . . 4
6.1.5.1 Unconfounded assignment . . . 4
6.1.5.2 The role of randomization . . . 5
6.1.5.3 Observational studies . . . 5
6.1.5.4 Identification of the ATE . . . 5
6.1.5.5 Regression estimation of the ATE . . . 6
6.1.5.6 Summary . . . 6
6.1.6 Appendix: Positivity and consistency in causal diagrams . . . 6
6.1.7 Appendix: Confounding and exchangeability . . . 6

6.2 Inverse probability weighting and doubly robust estimation **6**
6.2.1 The propensity score . . . 7
6.2.2 Inverse probability weighting . . . 7
6.2.2.1 The IPW identity . . . 7
6.2.2.2 Estimators . . . 7
6.2.2.3 Normalizing the weights . . . 8
6.2.2.4 The weighted population . . . 8
6.2.2.5 Pros and cons . . . 8
6.2.2.6 Trimming and truncation . . . 8
6.2.2.7 PS weighting in randomized trials . . . 8
6.2.2.8 Weighting versus balancing . . . 9
6.2.3 Doubly robust estimation . . . 9
6.2.3.1 Terminology . . . 9
6.2.3.2 Two building blocks . . . 9
6.2.3.3 Verification with the true nuisance functions . . . 9
6.2.3.4 Two equivalent forms . . . 10
6.2.3.5 What does it estimate? . . . 10
6.2.3.6 The doubly robust ATE estimator . . . 10
6.2.3.7 Concluding remarks . . . 10
6.2.3.8 Summary . . . 11

6.3 Conditional average treatment effects **11**
6.3.1 Why heterogeneous treatment effects? . . . 11
6.3.1.1 The average can be misleading . . . 11
6.3.1.2 Effect modification . . . 11

6.3.1.3 Why care? . . . 12
6.3.1.4 From stratification to CATE estimation . . . 12
6.3.2 Meta-learners for CATE . . . 12
6.3.2.1 T-learner and S-learner . . . 12
6.3.2.2 Regression imputation . . . 13
6.3.2.3 X-learner . . . 13
6.3.2.4 DR-learner . . . 13

6.4 The R-learner **14**
6.4.1 Motivation and identification . . . 14
6.4.2 The R-loss . . . 14
6.4.3 Algorithm . . . 15
6.4.4 Orthogonality and the quasi-oracle property . . . 15
6.4.5 Comparison and summary . . . 16

6.1 Potential outcomes

Motivation. The central question is how causal effects are defined and identified when causation is formulated as a comparison of potential outcomes, a missing-data problem, rather than through structural equations and graphs.

6.1.1 Frameworks for causal inference

Motivation. Place the potential outcome (PO) framework next to the causal-diagram framework.

A framework for causal inference must be rich enough to formalize basic intuitions about cause and effect. Two are in common use:

- the **potential outcome framework**, also called the counterfactual framework or Neyman–Rubin causal model (Neyman 1923; Rubin 1974);
- the **causal diagram framework** (Pearl 2009).

Mathematically the two are connected (Richardson and Robins 2013; see the SWIG lecture).

The PO framework is arguably the most widely used across medicine, health care, policy and the social sciences. Causal inference is viewed as a *missing-data problem*, with explicit modeling of the *assignment mechanism* as the process that reveals the observed data. It is rooted in Fisher's (1918, 1925) and Neyman's (1923) work on randomized experiments, extended by Rubin (1974, 1978) and many others to non-randomized studies.

6.1.2 Basic concepts

Motivation. Define units, treatments, potential outcomes and causal effects, and state the fundamental problem.

- **Unit:** the person, place or thing on which a treatment operates, at a particular time. (The same person at two different times constitutes two units.)
- **Treatment:** an intervention whose effect on some measurement of the units the investigator wishes to assess relative to no intervention (control).
- **Potential outcomes:** the values of a unit's measurement after application of the active treatment, $Y(1)$, and after non-application (control), $Y(0)$. Synonymously, *counterfactuals*.
- **Causal effect:** for each unit, a comparison of $Y(1)$ and $Y(0)$, e.g. $Y_i(1) - Y_i(0)$.

6.1.2.1 The fundamental problem of causal inference

At most one potential outcome is observed for each unit; the other is missing, i.e. counterfactual (Holland 1986).

Unit	$Y(x{=}0)$	$Y(x{=}1)$	X	Y
1	?	1	1	1
2	0	?	0	0
3	?	0	1	0
4	?	1	1	1
5	1	?	0	1

Neyman's urn model. Picture each unit as a ticket carrying both Y_1 and Y_0, placed in an urn. Assignment draws tickets and, for each, reveals only one half: Y_1 for treated units, Y_0 for controls.

6.1.2.2 God's eye view: individual versus average effects

Outcome Y: death (1) or survival (0). With both potential outcomes visible (adapted from Hernán and Robins, *What If*, Table 1.1):

	$Y(0)$	$Y(1)$	effect		$Y(0)$	$Y(1)$	effect
Rheia	1	0	-1	Hestia	1	0	-1
Zeus	0	0	0	Poseidon	1	0	-1
Kronos	0	1	$+1$	Hera	1	0	-1
Demeter	0	0	0	Artemis	1	1	0
Hades	0	0	0	Apollo	0	1	$+1$
Risk: $\mathbb{E}[Y(0)] = 0.5$, $\mathbb{E}[Y(1)] = 0.3$							

The average causal effect is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = 0.3 - 0.5 = -0.2$: treatment reduces risk. But individual effects vary: 4 helped (-1), 2 harmed ($+1$), 4 unaffected. We can never observe both columns for the same person.

6.1.2.3 What the researcher actually sees

(Adapted from *What If*, Table 1.2.)

	$Y(0)$	$Y(1)$	X	Y^{obs}		$Y(0)$	$Y(1)$	X	Y^{obs}
Rheia	?	0	1	0	Hestia	?	0	1	0
Zeus	0	?	0	0	Poseidon	1	?	0	1
Kronos	?	1	1	1	Hera	?	0	1	0
Demeter	0	?	0	0	Artemis	1	?	0	1
Hades	0	?	0	0	Apollo	?	1	1	1

The "?" entries are counterfactual and forever missing. By consistency $Y^{\text{obs}} = Y(X)$. The naive comparison gives $\bar{Y} = 2/5 = 0.4$ among the treated and $2/5 = 0.4$ among controls, a difference of $0 \neq -0.2$. Why? Treatment was not randomly assigned: sicker patients (those with $Y(0) = 1$) chose treatment more often (confounding). This is why we need exchangeability.

6.1.3 Assumptions linking potential and observed outcomes

Motivation. State SUTVA and consistency, and illustrate their failures.

6.1.3.1 SUTVA

Assumption 1 (Stable Unit Treatment Value Assumption; Rubin 1980). SUTVA comprises (1) *no interference* and (2) *no different versions* of a treatment (e.g. the same surgery performed by different surgeons). It connects the observed treatment X with the intervention of interest x:

$$X_i = 1 \Rightarrow Y_i = Y_i(1), \qquad X_i = 0 \Rightarrow Y_i = Y_i(0),$$

equivalently

$$Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

Violation by interference: $Y_i(x)$ depends on the treatments received by other units, e.g. vaccination, advertising, infectious diseases, social networks, agricultural plots. Then other assumptions are needed (Rosenbaum 2007; Hudgens and Halloran 2008), and spillover effects are often themselves of interest.

On the history of SUTVA (VanderWeele and Hernán, "Causal inference under multiple versions of treatment"). Rubin (1980) pointed out that the notation $Y_j(a)$ presupposes (i) that j's outcome under a does not depend on the treatments of other individuals and (ii) that there are no multiple versions of a leading to different outcomes. The first is "no interference" (attributed to Cox 1958), the second "no versions of treatment" (attributed to Neyman 1935). Also included in SUTVA is *consistency* (Robins 1986): $Y_j(a) = Y_j$ when $A_j = a$, tying counterfactual data to observed data. These assumptions matter in both experimental and observational studies; although usually stated only for observational research, they also govern the interpretation of randomized trials.

6.1.3.2 Consistency and well-defined interventions

Consistency: if $X_i = x$ then $Y_i = Y_i(x)$. It sounds obvious, but requires that "$X = x$" means the same thing for everyone. Example: the effect of "losing 20 kg" on heart-attack risk. Losing 20 kg through diet and exercise ($Y(x)$ = survived) and losing it through starvation or illness ($Y(x)$ = died) are the same nominal treatment with different outcomes, so $Y(x)$ is not well-defined. Fix: specify the intervention precisely ("20 kg loss via a supervised diet and exercise program"). Vague treatments give vague causal questions and meaningless answers (*What If*, Fine Point 3.4).

6.1.4 Causal estimands

Motivation. List the quantities typically targeted.

Differences.

$$\begin{aligned} \text{ATE:} \quad & \tau = \mathbb{E}[Y(1) - Y(0)], \\ \text{CATE:} \quad & \tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z], \\ \text{ATT:} \quad & \text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid X = 1]. \end{aligned}$$

Ratios. Individual ratio $r_i = Y_i(1)/Y_i(0)$; population ratio $r = \mathbb{E}[Y(1)]/\mathbb{E}[Y(0)]$; relative effect (lift) $l = \mathbb{E}[Y(1) - Y(0)]/\mathbb{E}[Y(0)]$. Note that the population ratio is not an average of individual ratios unless the effect is homogeneous. Continuous outcomes (income, test scores) dominate in the social sciences; binary outcomes (mortality, readmission) are more common in medicine and health.

6.1.5 Randomization and observational studies

Motivation. Show why randomization identifies causal effects and which assumptions replace it in observational data.

6.1.5.1 Unconfounded assignment

An assignment mechanism is *unconfounded* if it does not depend on the potential outcomes, only on pre-treatment covariates:

$$p(X = 1 \mid Z, Y(0), Y(1)) = p(X = 1 \mid Z).$$

$p(X = 1 \mid Z)$ is the *propensity score* (Rosenbaum and Rubin 1983). In randomized experiments the mechanism is unconfounded, and moreover known and controlled by the investigator.

6.1.5.2 The role of randomization

Randomization balances observed covariates ($X \perp\!\!\!\perp Z$), unobserved covariates ($X \perp\!\!\!\perp U$) and potential outcomes, i.e. it guarantees ignorability

$$X \perp\!\!\!\perp (Y(1), Y(0)).$$

("Covariates", "pre-treatment variables" and "baseline characteristics" are used interchangeably for Z.) Under randomization causal effects are identified:

$$P(Y(x)) = P(Y(x) \mid X = x) = P(Y \mid X = x), \qquad x \in \{0, 1\},$$

the first equality by ignorability and the second by consistency. Association then implies causation (within the PO framework and its assumptions):

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid X = 1] - \mathbb{E}[Y(0) \mid X = 0] = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0].$$

As graphs. In a randomized experiment nothing points into X (e.g. $X \to Y \leftarrow Z$). A single-world intervention graph (SWIG) splits the treatment node into the observed X and the intervened value x, making potential outcomes explicit: $X \mid x \to Y(x) \leftarrow Z$. Since X and $Y(x)$ are d-separated, $Y(x) \perp\!\!\!\perp X$.

6.1.5.3 Observational studies

The most common assumptions:

1. **No unmeasured confounding:** $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$. Also called unconfoundedness, ignorability, selection on observables, no omitted variables, exogeneity, conditional exchangeability. Conditional on covariates, X_i is effectively randomly assigned.
2. **Positivity (overlap):** $0 < P(X_i = 1 \mid Z_i) < 1$; both treatment and control are possible at every value of Z_i.

We take Z as given for now; choosing it is exactly the adjustment-criterion problem. These assumptions can be wrong. Graphically, $Z \to X$, $Z \to Y$, $X \to Y$; in the SWIG, $Z \to X$, $Z \to Y(x)$, $x \to Y(x)$, and $Y(x) \perp\!\!\!\perp X \mid Z$ is read off by d-separation.

6.1.5.4 Identification of the ATE

Under positivity and no unmeasured confounding,

$$\begin{aligned}
\tau &= \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_Z\{\mathbb{E}[Y(1) - Y(0) \mid Z]\} = \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid Z] - \mathbb{E}[Y(0) \mid Z]\} \\
&= \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid X = 1, Z] - \mathbb{E}[Y(0) \mid X = 0, Z]\} && \text{(ignorability)} \\
&= \mathbb{E}_Z\{\mathbb{E}[Y \mid X = 1, Z] - \mathbb{E}[Y \mid X = 0, Z]\} && \text{(consistency).}
\end{aligned}$$

Positivity ensures that the conditional expectations in the last line are defined for every z. Define the treated and control conditional expectation functions (CEFs)

$$\mu_1(z) = \mathbb{E}[Y(1) \mid Z = z], \qquad \mu_0(z) = \mathbb{E}[Y(0) \mid Z = z],$$

describing how the means of the potential outcomes vary with covariates. The key step of the proof is

$$\underbrace{\mu_1(z)}_{\text{counterfactual}} = \underbrace{\mathbb{E}[Y \mid X = 1, Z = z]}_{\text{observational}}, \qquad \mu_0(z) = \mathbb{E}[Y \mid X = 0, Z = z].$$

6.1.5.5 Regression estimation of the ATE

Fit regressions $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$ (linear or non-linear); the safest practice is to estimate separate regressions in each treatment group. Then

$$\hat{\tau}_{\text{reg}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\right).$$

Procedure: predict for all units with $X_i = 1$; predict for all units with $X_i = 0$; average the differences.

6.1.5.6 Summary

Potential outcomes	$Y(1), Y(0)$
Conditional ignorability (back-door)	$Y(1), Y(0) \perp\!\!\!\perp X \mid Z$

Identification: $\mathbb{E}[Y(1)] = \mathbb{E}[\mathbb{E}[Y(1) \mid Z]] = \mathbb{E}[\mathbb{E}[Y(1) \mid X = 1, Z]] = \mathbb{E}[\mathbb{E}[Y \mid X = 1, Z]]$.

6.1.6 Appendix: Positivity and consistency in causal diagrams

Positivity concerns arrows into the treatment node. If X is a deterministic function of a pre-treatment Z (a bold/deterministic edge $Z \to X$), positivity is violated by design. In causal diagrams positivity is implicit unless otherwise specified.

Consistency concerns arrows out of the treatment node. $X \to Y$ must correspond to a well-defined intervention. If $X = x$ can be achieved in several ways with different effects on Y (weight loss via diet, exercise or surgery), the effect is ill-defined. Treatment nodes thus have a special status: they must correspond to sufficiently well-defined interventions (*What If*, §6.4).

6.1.7 Appendix: Confounding and exchangeability

- **Marginal exchangeability** $Y(x) \perp\!\!\!\perp X$ holds when X and Y have no common causes (no open back-door paths), as in a marginally randomized experiment ($X \to Y$ only).
- **Conditional exchangeability** $Y(x) \perp\!\!\!\perp X \mid Z$ holds when all back-door paths are blocked by Z, as in a conditionally randomized experiment ($Z \to X \to Y$, $Z \to Y$).
- If Z satisfies the back-door criterion, conditional exchangeability holds. The converse requires the more general adjustment criterion (Perković et al. 2018).
- Confounding (graphical, structural) = lack of exchangeability (counterfactual) (*What If*, §7.2 and Technical Point 7.1).

6.2 Inverse probability weighting and doubly robust estimation

Motivation. The central question is how to estimate $\mathbb{E}[Y(x)]$ and the ATE from observational data once unconfoundedness and positivity are granted, and how to protect the estimate against misspecification of the models we must fit.

Recall the key assumptions of the PO framework:

(1) unconfoundedness $\{Y(1), Y(0)\} \perp\!\!\!\perp X \mid Z$;

(2) positivity $0 < P(X = 1 \mid Z) < 1$;

(3) hence identification by adjustment, $\mathbb{E}[Y(x)] = \mathbb{E}_Z[\mathbb{E}[Y \mid X = x, Z]]$.

6.2.1 The propensity score

Motivation. Reduce the covariate vector to a scalar that suffices for confounding control.

Definition (Propensity score; Rosenbaum and Rubin 1983)

$e(Z) = P(X = 1 \mid Z) = \mathbb{E}[X \mid Z]$, where $Z = (Z_1, \ldots, Z_p)$ collects the p pre-treatment covariates.

Proposition 6.1 (Property 1: balancing)

$X \perp\!\!\!\perp Z \mid e(Z)$. More generally a *balancing score* is any $b(Z)$ with $X \perp\!\!\!\perp Z \mid b(Z)$; $e(Z)$ is one (Z itself trivially is another), and Rosenbaum and Rubin (1983) show that $e(Z)$ is the *coarsest* balancing score.

Proof [Supplement]. It suffices to show $P(X = 1 \mid Z, e(Z)) = P(X = 1 \mid e(Z))$. The left side is $P(X = 1 \mid Z) = e(Z)$ since $e(Z)$ is a function of Z. The right side is $\mathbb{E}[\mathbb{E}[X \mid Z] \mid e(Z)] = \mathbb{E}[e(Z) \mid e(Z)] = e(Z)$. ■

Proposition 6.2 (Property 2: unconfoundedness given the PS)

If $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$, then $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid e(Z_i)$.

Proof [Supplement]. $P(X = 1 \mid Y(1), Y(0), e(Z)) = \mathbb{E}\big[P(X = 1 \mid Y(1), Y(0), Z) \mid Y(1), Y(0), e(Z)\big] = \mathbb{E}\big[e(Z) \mid Y(1), Y(0), e(Z)\big] = e(Z)$, which does not depend on the potential outcomes. ■

Given covariates that ensure unconfoundedness, adjusting for differences in the propensity score removes all biases associated with differences in the covariates; $e(Z)$ is a summary score of the observed covariates.

6.2.2 Inverse probability weighting

Motivation. Estimate potential-outcome means by reweighting observed outcomes.

6.2.2.1 The IPW identity

$$\text{ATE} = \mathbb{E}\left[\frac{XY}{e(Z)} - \frac{(1-X)Y}{1-e(Z)}\right].$$

Using consistency ($XY = XY(1)$) and unconfoundedness ($X \perp\!\!\!\perp Y(1) \mid Z$),

$$\mathbb{E}\left[\frac{XY}{e(Z)}\right] = \mathbb{E}\left[\frac{\mathbb{E}[XY(1) \mid Z]}{e(Z)}\right] \overset{\text{uncf.}}{=} \mathbb{E}\left[\frac{\mathbb{E}[X \mid Z]\,\mathbb{E}[Y(1) \mid Z]}{e(Z)}\right] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid Z]\big] = \mathbb{E}[Y(1)],$$

and symmetrically for the control term. This is the Horvitz–Thompson (HT) estimator of survey sampling, also called inverse probability of treatment weighting (IPTW).

6.2.2.2 Estimators

With an estimated PS $\hat{e}$, define the weights $\hat{w}_1(Z_i) = 1/\hat{e}(Z_i)$ for $X_i = 1$ and $\hat{w}_0(Z_i) = 1/(1-\hat{e}(Z_i))$ for $X_i = 0$. The difference of weighted means

$$\hat{\tau}_{\text{ipw},1} = \frac{1}{N}\left(\sum_{i=1}^{N} \frac{Y_i X_i}{\hat{e}(Z_i)} - \sum_{i=1}^{N} \frac{Y_i(1-X_i)}{1-\hat{e}(Z_i)}\right) = \frac{1}{N}\sum_{i=1}^{N} \{Y_i X_i \hat{w}_1(Z_i) - Y_i(1-X_i)\hat{w}_0(Z_i)\}$$

is unbiased when $\hat{e} = e$.

6.2.2.3 Normalizing the weights

Since $\mathbb{E}\big[\mathbb{E}[X \mid Z]/e(Z)\big] = 1$, the weights average to one in expectation, but $\frac{1}{N}\sum_i X_i/\hat{e}(Z_i)$ need not equal one in a sample. Good practice for any weighting method is to normalize so that the weights within each group sum to one: divide $\hat{w}_i$ by $\sum_{j:X_j=x} \hat{w}_j$. This gives the Hájek estimator ("normalized" or "self-normalized" weights):

$$\hat{\tau}_{\text{ipw},2} = \frac{\sum_i Y_i X_i \hat{w}_1(Z_i)}{\sum_i X_i \hat{w}_1(Z_i)} - \frac{\sum_i Y_i (1 - X_i) \hat{w}_0(Z_i)}{\sum_i (1 - X_i) \hat{w}_0(Z_i)}.$$

It typically reduces variance, $V(\hat{\tau}_{\text{ipw},2}) \leq V(\hat{\tau}_{\text{ipw},1})$, and gives more stable estimates (Hirano, Imbens and Ridder 2003).

6.2.2.4 The weighted population

IPW creates a weighted (target) population that the sample represents. In it, covariate distributions are balanced between groups:

$$\mathbb{E}\left[\frac{X}{e(Z)} Z\right] = \mathbb{E}\left[\frac{1 - X}{1 - e(Z)} Z\right] \ (= \mathbb{E}[Z]),$$

since $\mathbb{E}[XZ/e(Z)] = \mathbb{E}[\mathbb{E}[X \mid Z]Z/e(Z)] = \mathbb{E}[Z]$. Intuition: the less likely a subject was to be treated, the larger the population it should represent.

6.2.2.5 Pros and cons

Advantages: simple, with a theoretical foundation (Horvitz–Thompson); model-free for the outcome; explicit target population (ATE, ATT, ...) via the choice of weights; extends to complex settings (time-varying treatments, marginal structural models).
Disadvantages: more sensitive to misspecification of the PS; propensity scores near 0 or 1 yield extreme weights, handled by many ad hoc decisions.

6.2.2.6 Trimming and truncation

- **Symmetric trimming** (Crump et al. 2009): exclude units whose estimated PS lies outside $[\alpha, 1 - \alpha]$; rule of thumb $\alpha = 0.1$.
- **Asymmetric trimming** (Stürmer et al. 2010): exclude units outside the common PS range of treated and controls; then further exclude units whose PS is below the q quantile of the treated, and units whose PS is above the $(1 - q)$ quantile of the controls.
- **Truncation:** set estimated PS below α to α and above $1 - \alpha$ to $1 - \alpha$.

Picture: the PS histograms of untreated (left-heavy) and treated (right-heavy) units have non-overlapping tails; range restriction removes the lowest percentiles of the treated and the highest percentiles of the untreated.

6.2.2.7 PS weighting in randomized trials

Although developed for observational studies, PS weighting can be used for covariate adjustment in RCTs: chance imbalances are common, and adjustment improves precision. The procedure is the same as in observational studies (with an estimated, rather than the known, PS).

6.2.2.8 Weighting versus balancing

IPW balances covariates *in expectation* but not in finite samples. A recent line of work finds weights w_i directly so that weighted covariate means are (approximately) balanced, bypassing PS estimation:

$$\frac{1}{N_1}\sum_{i=1}^{N} w_i X_i Z_i \approx \frac{1}{N_0}\sum_{i=1}^{N} w_i (1-X_i) Z_i,$$

typically minimizing a measure of weight dispersion (e.g. entropy) subject to balance constraints, by numerical optimization. Examples: entropy balancing (Hainmueller 2011), covariate balancing propensity score (Imai and Ratkovic 2014), stabilized balancing weights (Zubizarreta 2015). These balance only the functions (moments) of Z the analyst specifies. If the true outcome model is $Y \sim Z_1 + Z_2 + Z_1 Z_2$ and we balance $Z_1, Z_2, Z_1^2, Z_2^2, \ldots$ but not $Z_1 Z_2$, the interaction may be severely imbalanced and the result biased.

6.2.3 Doubly robust estimation

Motivation. Combine outcome regression and IPW so that consistency requires only one of the two models to be correct.

6.2.3.1 Terminology

- **Correctly specified model:** the true function belongs to the assumed class (e.g. $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$ fitted with a linear model with interaction).
- **Misspecified model:** it does not (e.g. the truth is non-linear in Z but a linear model is fitted). The fit converges to the best approximation within the class, which is generally biased.
- **Consistent estimator:** $\hat{\theta}_N \xrightarrow{p} \theta$ as $N \to \infty$. Correct specification gives consistency (under regularity conditions); misspecification generally does not.

6.2.3.2 Two building blocks

Since $\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$ is symmetric under $X \leftrightarrow 1-X$, $e \leftrightarrow 1-e$, focus on $\mu_1 = \mathbb{E}[Y(1)]$. Let $m_x(Z) = \mathbb{E}[Y(x) \mid Z]$ (written $\mu_x(z)$ in Section 6.1).

- **Outcome regression:** $\hat{\mu}_{1,\text{adj}} = N^{-1}\sum_i \hat{m}_1(Z_i)$; consistent if m_1 is correctly specified.
- **IPW:** $\hat{\mu}_{1,\text{ipw}} = N^{-1}\sum_i Y_i X_i / \hat{e}(Z_i)$; consistent if the PS model (e.g. logistic) is correct, but can be inefficient.

Can the virtues be combined? Yes:

$$\hat{\mu}_{1,\text{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right].$$

6.2.3.3 Verification with the true nuisance functions

$$\mathbb{E}\left[\frac{XY}{e(Z)} - \frac{X-e(Z)}{e(Z)}m_1(Z)\right] = \mathbb{E}\left[\frac{XY - X m_1(Z) + e(Z)m_1(Z)}{e(Z)}\right] = \mathbb{E}\left[m_1(Z) + \frac{X}{e(Z)}\{Y - m_1(Z)\}\right] = \mu_1,$$

since by iterated expectations and $\mathbb{E}[X \mid Z] = e(Z)$ (as in the IPW derivation) $\mathbb{E}[X\{Y(1) - m_1(Z)\}/e(Z)] = \mathbb{E}[\mathbb{E}[Y(1) - m_1(Z) \mid Z]] = 0$.

6.2.3.4 Two equivalent forms

$$\hat{\mu}_{1,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right] \qquad \text{(IPW augmented by OR)}$$

$$= \frac{1}{N}\sum_{i=1}^{N}\left[\hat{m}_1(Z_i) + \frac{X_i\{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}(Z_i)}\right] \qquad \text{(OR augmented by IPW residual correction).}$$

The forms are algebraically identical but suggest different statistical interpretations.

6.2.3.5 What does it estimate?

With arbitrary limits $\hat{e}$, $\hat{m}_1$, using $XY = XY(1)$:

$$\mathbb{E}\left[\frac{XY}{\hat{e}(Z)} - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right] = \mathbb{E}\left[Y(1) + \frac{X - \hat{e}(Z)}{\hat{e}(Z)}Y(1) - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right]$$

$$= \mathbb{E}[Y(1)] + \underbrace{\mathbb{E}\left[\frac{X - \hat{e}(Z)}{\hat{e}(Z)}\{Y(1) - \hat{m}_1(Z)\}\right]}_{R}.$$

$\hat{\mu}_{1,\mathrm{dr}}$ estimates $\mathbb{E}[Y(1)]$ iff $R = 0$: a regression estimator augmented by weighted residuals.

Scenario 1: PS wrong, OR right ($\hat{m}_1 = m_1$). Conditioning on Z and using unconfoundedness,

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right], \qquad \mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z] = m_1(Z) - \hat{m}_1(Z) = 0.$$

Scenario 2: PS right, OR wrong ($\hat{e} = e$).

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right] = 0 \quad \text{since } e(Z) - \hat{e}(Z) = 0.$$

In both cases $R = 0$ and $\hat{\mu}_{1,\mathrm{dr}}$ is consistent. By the same argument with $X \to 1 - X$, $e \to 1 - e$, $m_1 \to m_0$,

$$\hat{\mu}_{0,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]$$

is consistent for $\mathbb{E}[Y(0)]$, and if both models are correct it is of course consistent too.

6.2.3.6 The doubly robust ATE estimator

$$\hat{\tau}_{\mathrm{dr}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right]}_{\hat{\mu}_{1,\mathrm{dr}}}$$

$$- \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]}_{\hat{\mu}_{0,\mathrm{dr}}}.$$

6.2.3.7 Concluding remarks

- **Double robustness:** $\hat{\tau}_{\mathrm{dr}}$ is consistent if either the PS model or the outcome model (not necessarily both) is correctly specified: two chances to get it right. DR is a large-sample property.

- If both e and m_x are correct, $\hat{\tau}_{\text{dr}}$ has smaller asymptotic variance than IPW. If the outcome model is correct, it has larger asymptotic variance than the direct regression estimator.
- Empirically the outcome model matters more: misspecifying $\hat{m}$ causes much larger bias than misspecifying $\hat{e}$, and even with a correct $\hat{e}$ a misspecified $\hat{m}$ usually gives notable bias; the extra protection from PS augmentation is somewhat limited.
- Kang and Schafer (2007, *Stat. Sci.*) give an example where moderate misspecification of both models (a likely scenario) leads to large bias and variance of DR.

6.2.3.8 Summary

1. The propensity score $e(Z) = P(X = 1 \mid Z)$ is a scalar summary of covariates that balances treatment groups.
2. IPW reweights by $1/e(Z)$ to remove confounding. Pros: simple, outcome-model-free. Cons: extreme weights.
3. Outcome regression models $\mathbb{E}[Y \mid X, Z]$ directly. Pros: efficient when correct. Cons: sensitive to misspecification.
4. Doubly robust estimation combines both and is consistent if either model is correct.

6.3 Conditional average treatment effects

Motivation. The central question is how treatment effects vary with covariates, why that matters, and how flexible machine-learning "meta-learners" estimate the conditional average treatment effect $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$.

6.3.1 Why heterogeneous treatment effects?

Motivation. Show that a single average can hide opposite effects in subgroups.

6.3.1.1 The average can be misleading

Consider 20 individuals, half women ($Z = 1$) and half men ($Z = 0$), with a null average effect:

$$P(Y(1) = 1) - P(Y(0) = 1) = 0.5 - 0.5 = 0.$$

Yet among women the causal risk difference is $0.6 - 0.4 = +0.2$ (treatment increases the risk of death) and among men $0.4 - 0.6 = -0.2$ (it decreases it). Indeed $\frac{1}{2}(+0.2) + \frac{1}{2}(-0.2) = 0$. *There is generally no such thing as "the causal effect of treatment"; it depends on the characteristics of the population studied* (Hernán and Robins, *What If*, §4.1).

6.3.1.2 Effect modification

Definition

Z is a *modifier* of the effect of X on Y when the average causal effect of X on Y varies across levels of Z.

- **Additive** effect modification: $\mathbb{E}[Y(1) - Y(0) \mid Z = 1] \neq \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$.

- **Multiplicative** effect modification: $\dfrac{\mathbb{E}[Y(1) \mid Z=1]}{\mathbb{E}[Y(0) \mid Z=1]} \neq \dfrac{\mathbb{E}[Y(1) \mid Z=0]}{\mathbb{E}[Y(0) \mid Z=0]}$.

Modification may appear on one scale but not on the other (e.g. risks $0.2 \to 0.4$ and $0.4 \to 0.8$ have equal ratios but different differences). *Qualitative* effect modification (effects in opposite directions across strata) implies modification on both scales.

6.3.1.3 Why care?

1. **Policy targeting:** identify who benefits. If treatment harms women and helps men (qualitative modification), a physician would, absent other information, treat only men.
2. **Transportability:** if the distribution of modifiers (sex, age) differs between the study and a target population, so does the ATE.
3. **Mechanisms:** modification may reveal biological, social or other mechanisms.

Conditioning on the full covariate vector gives the CATE, $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$ (*What If*, §4.3).

6.3.1.4 From stratification to CATE estimation

The traditional approach computes the effect within each stratum of Z and compares stratum-specific measures. Its limitations: continuous modifiers must be discretized; the number of strata grows exponentially with the number of covariates (curse of dimensionality); strata may contain too few observations. Modern CATE estimators (meta-learners, causal forests) use flexible ML models to estimate $\tau(z)$ without explicit stratification (*What If*, §4.2).

6.3.2 Meta-learners for CATE

Motivation. Build CATE estimators from arbitrary regression algorithms (Künzel et al. 2019, PNAS).

Under unconfoundedness and positivity, $\mu_x(z) := \mathbb{E}[Y(x) \mid Z = z] = \mathbb{E}[Y \mid X = x, Z = z]$, so any regression method can be used.

6.3.2.1 T-learner and S-learner

T-learner (two models): fit $\hat{\mu}_1(z)$ on the treated and $\hat{\mu}_0(z)$ on the controls separately;

$$\hat{\tau}_T(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z), \qquad \widehat{\mathrm{ATE}}_T = \frac{1}{n}\sum_{i=1}^{n} \big(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\big).$$

S-learner (single model): fit one model $\hat{\mu}(z, x) \approx \mathbb{E}[Y \mid Z = z, X = x]$ with X as an ordinary feature;

$$\hat{\tau}_S(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0), \qquad \widehat{\mathrm{ATE}}_S = \frac{1}{n}\sum_{i=1}^{n} \big(\hat{\mu}(Z_i, 1) - \hat{\mu}(Z_i, 0)\big).$$

The T-learner can be noisy when one arm is small; the S-learner may shrink the effect of X towards zero if the learner regularizes it away.

6.3.2.2 Regression imputation

Plug in $\hat{\mu}$ only for the *unobserved* counterfactual and use Y_i for the observed one:

$$\hat{\tau}_{RI} = \frac{1}{n}\sum_{i=1}^{n}\Big[\big(X_iY_i + (1-X_i)\hat{\mu}_1(Z_i)\big) - \big((1-X_i)Y_i + X_i\hat{\mu}_0(Z_i)\big)\Big].$$

For treated units ($X_i = 1$): observed Y_i for $Y(1)$, imputed $\hat{\mu}_0(Z_i)$ for $Y(0)$; for controls: imputed $\hat{\mu}_1(Z_i)$ for $Y(1)$, observed Y_i for $Y(0)$.

6.3.2.3 X-learner

1. Estimate the response functions $\hat{\mu}_0(z) \approx \mathbb{E}[Y(0) \mid Z = z]$ and $\hat{\mu}_1(z) \approx \mathbb{E}[Y(1) \mid Z = z]$.
2. Impute individual treatment effects: for the treated $\tilde{D}_i^1 := Y_i - \hat{\mu}_0(z_i)$; for the controls $\tilde{D}_i^0 := \hat{\mu}_1(z_i) - Y_i$. Regress to obtain $\hat{\tau}_1(z) \approx \mathbb{E}[\tilde{D}_i^1 \mid Z = z]$ (from the treated) and $\hat{\tau}_0(z) \approx \mathbb{E}[\tilde{D}_i^0 \mid Z = z]$ (from the controls).
3. Combine with a weight function $g(z) \in [0,1]$, e.g. $g(z) = \hat{e}(z)$:

$$\hat{\tau}(z) = g(z)\,\hat{\tau}_0(z) + (1 - g(z))\,\hat{\tau}_1(z).$$

Illustration. In a simulation (one covariate $x \in [-1,1]$) the true effect is $\tau = 1$ and treated units are rare ($g = \hat{e} \approx 0.05$). Step 1 fits $\hat{\mu}_0$ as a piecewise-linear function (many control data) and $\hat{\mu}_1$ as a linear function (few treated data). $\hat{\mu}_0$, fitted on the large control group, is flexible and contains jumps; $\hat{\mu}_1$, fitted on few treated points, is nearly linear. The T-learner $\hat{\tau}_T = \hat{\mu}_1 - \hat{\mu}_0$ inherits the jumps of $\hat{\mu}_0$, while the X-learner $\hat{\tau}_X \approx (1-g)\hat{\tau}_1$ is nearly linear and close to the true τ: $\hat{\tau}_1$ uses the imputed effects of the treated, computed against the accurate $\hat{\mu}_0$, and is smooth because the effect itself is simple.

6.3.2.4 DR-learner

Apply the doubly robust score of Section 6.2 to learn $\tau(z)$:

1. Estimate the nuisances $\hat{\mu}_x(z)$ and $\hat{e}(z)$.
2. Construct the pseudo-outcome

$$\tilde{Y}_i = \frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1-X_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{e}(Z_i)} + \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i).$$

3. Regress $\tilde{Y}_i$ on Z_i with any flexible method to obtain $\hat{\tau}(z)$.

Why the DR-learner?

- $\mathbb{E}[\tilde{Y}_i \mid Z_i = z] = \tau(z)$ when the nuisances are correct (by the same computation as for the DR estimator, conditionally on Z); in fact one correct nuisance suffices.
- *Neyman orthogonality:* errors in $\hat{\mu}, \hat{e}$ affect $\hat{\tau}$ only at second order (through products of the two errors).
- It adapts to the complexity of $\tau(z)$, even when the nuisance functions are more complex (Kennedy 2023).
- In practice use K-fold cross-fitting: estimate the nuisances on folds $\neq k$ and construct $\tilde{Y}_i$ on fold k, to avoid overfitting bias.

6.4 The R-learner

Motivation. The central question is how to estimate the conditional average treatment effect with a loss function whose minimizer is the CATE itself, so that errors in the nuisance regressions affect the estimate only at second order.

6.4.1 Motivation and identification

Motivation. Set up the target and the residualised representation.

We observe i.i.d. $O_i = (X_i, Z_i, Y_i)$ with $X_i \in \{0, 1\}$ and want $\tau(z) = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = z]$. S-, T- and X-learners are easy to explain but can pass nuisance-regression bias into $\hat{\tau}$. DR-learners build a pseudo-outcome from a Neyman-orthogonal DR score and regress it on Z. The R-learner instead builds a *loss function whose target is $\tau(\cdot)$ itself* (Nie and Wager 2021).

Assumptions: consistency $Y_i = Y_i(X_i)$, unconfoundedness $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i \mid Z_i$, overlap $0 < e(Z_i) < 1$. **Nuisances:**

$$e(z) = \Pr(X = 1 \mid Z = z), \qquad m(z) = \mathbb{E}[Y \mid Z = z],$$

where m is the Z-only outcome mean, not an arm-specific mean μ_x. Residuals: $\tilde{Y}_i = Y_i - m(Z_i)$, $\tilde{X}_i = X_i - e(Z_i)$. The R-learner uses

$$\tilde{Y}_i = \tilde{X}_i \, \tau(Z_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$

Robinson's partially linear decomposition with a varying coefficient.

Derivation. Fix $Z = z$ and let $\mu_x(z) = \mathbb{E}[Y(x) \mid Z = z]$, $\tau(z) = \mu_1(z) - \mu_0(z)$. By consistency and unconfoundedness, $\mathbb{E}[Y \mid X = x, Z = z] = \mu_0(z) + x\,\tau(z)$. Since m averages over the observed treatment mix,

$$m(z) = \mathbb{E}[Y \mid Z = z] = e(z)\mu_1(z) + \{1 - e(z)\}\mu_0(z) = \mu_0(z) + e(z)\tau(z),$$

so $\mathbb{E}[Y - m(Z) \mid X, Z = z] = \{X - e(z)\}\tau(z)$, i.e. $\mathbb{E}[\tilde{Y} \mid X, Z = z] = \tilde{X}\tau(z)$; the remainder is ε.

6.4.2 The R-loss

Motivation. Turn the representation into a loss and interpret its minimizer.

Oracle R-loss. Since $\mathbb{E}[\tilde{Y} \mid X, Z] = \tilde{X}\tau(Z)$, an oracle knowing m and e could estimate τ by minimizing

$$L_R(\tau) = \mathbb{E}\big[\{\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\}^2\big], \qquad \tilde{\tau} = \arg\min_{\tau} \Big\{ \frac{1}{n} \sum_{i=1}^{n} \big(\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\} \Big\},$$

with a penalty/regularizer Λ_n (Robinson 1988; Nie and Wager 2021, §2).

A residual regression. Conditionally on $Z = z$ the oracle loss is $Q_z(t) = \mathbb{E}[(\tilde{Y} - \tilde{X}t)^2 \mid Z = z]$, a least-squares regression of $\tilde{Y}$ on $\tilde{X}$ through the origin. Its slope solves $0 = \mathbb{E}[\tilde{X}(\tilde{Y} - \tilde{X}t_z) \mid Z = z]$, so

$$t_z = \frac{\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z]}{\mathbb{E}[\tilde{X}^2 \mid Z = z]}.$$

Why this slope is the CATE. By iterated expectations, $\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z] = \mathbb{E}[\tilde{X}\,\mathbb{E}[\tilde{Y} \mid X, Z = z] \mid Z = z] = \mathbb{E}[\tilde{X}^2 \mid Z = z]\tau(z)$, hence $t_z = \tau(z)$. *The CATE is the population slope of outcome residuals on treatment residuals.*

6.4.3 Algorithm

Motivation. Implement the R-learner with cross-fitting.

1. Split the data into K folds (cross-fitting: each unit's residual uses nuisance models trained on other folds).
2. For each fold k, fit nuisances on the other folds: $\hat{m}^{(-k)}(z) \approx \mathbb{E}[Y \mid Z = z]$, $\hat{e}^{(-k)}(z) \approx \Pr(X = 1 \mid Z = z)$.
3. Out-of-fold residuals: $Y_i^r = Y_i - \hat{m}^{(-k_i)}(Z_i)$, $X_i^r = X_i - \hat{e}^{(-k_i)}(Z_i)$.
4. Estimate $\hat{\tau} = \arg\min_\tau \left\{ \frac{1}{n} \sum_{i=1}^n \left(Y_i^r - X_i^r \tau(Z_i)\right)^2 + \Lambda_n\{\tau(\cdot)\} \right\}$.

Workflow: data $(X, Z, Y) \to$ cross-fit $\hat{m}, \hat{e} \to$ residualise $Y - \hat{m}(Z)$, $X - \hat{e}(Z) \to$ minimize the R-loss $\to \hat{\tau}(z)$. First-stage prediction creates residualised signals; the second stage learns τ.

The R-loss weights residual treatment variation.

$$\sum_i \left(Y_i^r - X_i^r \tau(Z_i)\right)^2 = \sum_i (X_i^r)^2 \Big(\frac{Y_i^r}{X_i^r} - \tau(Z_i)\Big)^2.$$

Units with large $|X_i^r|$ (treatment poorly predicted by Z_i) carry information about $\tau(Z_i)$; units with small $|X_i^r|$ (treatment nearly determined by Z_i) put little pressure on the local $\tau(Z_i)$. R-learners learn the CATE where residual treatment variation remains.

What can optimize the R-loss? The second stage is empirical risk minimization, $\min_{\tau \in \mathcal{F}} \frac{1}{n} \sum_i (Y_i^r - X_i^r \tau(Z_i))^2 + \Lambda_n\{\tau\}$: simple choices (linear or sparse linear $\tau(z)$, splines, kernels) or flexible ones (boosting, random forests, neural networks, any learner with a squared-loss interface, using weights $(X_i^r)^2$ and targets Y_i^r / X_i^r). The learner for τ is tuned by *validation R-loss*, not by outcome prediction loss.

6.4.4 Orthogonality and the quasi-oracle property

Motivation. Show why nuisance errors matter only at second order.

A population moment is an expectation equation that is zero in the target population (before finite-sample fluctuation). For a candidate τ, use

$$\psi(O; \tau, m, e) = \{Y - m(Z) - (X - e(Z))\tau(Z)\}\{X - e(Z)\}, \qquad \mathbb{E}[\psi(O; \tau^*, m, e) \mid Z = z] = 0 \text{ at the truth.}$$

Proposition 6.3 (Neyman orthogonality)

Let $\hat{m} = m + \delta_m$, $\hat{e} = e + \delta_e$ and hold $\tau = \tau^*$. Then

$$\mathbb{E}[\psi(O; \tau^*, \hat{m}, \hat{e}) \mid Z = z] = \underbrace{\delta_m(z)\delta_e(z)}_{\text{product}} - \underbrace{\tau^*(z)\delta_e(z)^2}_{\text{square}}.$$

Proof [Supplement]. With $\tilde{Y} = Y - m(Z)$, $\tilde{X} = X - e(Z)$, the moment equals $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^* - \delta_m + \delta_e\tau^*)(\tilde{X} - \delta_e) \mid Z]$. The term $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^*)\tilde{X} \mid Z] = \mathbb{E}[\varepsilon \tilde{X} \mid Z] = 0$; the term $-\delta_e \mathbb{E}[\tilde{Y} - \tilde{X}\tau^* \mid Z] = -\delta_e \mathbb{E}[\varepsilon \mid Z] = 0$; and $(-\delta_m + \delta_e\tau^*)\mathbb{E}[\tilde{X} - \delta_e \mid Z] = (-\delta_m + \delta_e\tau^*)(-\delta_e)$ since $\mathbb{E}[\tilde{X} \mid Z] = 0$. ■

No first-order δ_m or δ_e term remains: small nuisance errors create only second-order bias, so flexible nuisance fits are usable (Nie and Wager 2021; Chernozhukov et al. 2018).

The quasi-oracle property. Measure excess risk with the oracle population R-risk $R(\tau) = \mathbb{E}[\{Y - m(Z) - (X - e(Z))\tau(Z)\}^2]$, via $R(\tau) - R(\tau^*)$. Let $\hat{\tau}$ fit residuals from $\hat{m}, \hat{e}$ and $\tilde{\tau}$ the oracle

version with true m, e. Nie and Wager show

$$R(\hat{\tau}) - R(\tau^*) \approx R(\tilde{\tau}) - R(\tau^*),$$

provided the nuisances are estimated well enough (RMSE $o(n^{-1/4})$, so their product is $o(n^{-1/2})$). The rate depends mainly on the complexity of τ: after residualisation, the second stage learns τ, not the full complexity of m and e.

Why cross-fitting matters. If observation i helps train $\hat{m}, \hat{e}$, then $\hat{m}(Z_i), \hat{e}(Z_i)$ can memorize noise in (Y_i, X_i): residuals become too small, the noise fitted by the nuisances leaks into the R-loss, and the second stage may mistake first-stage overfitting artefacts for heterogeneity in $\tau(Z)$. With cross-fitting, residuals are out-of-fold predictions, unit i's noise was not used to train its nuisances, and orthogonality can remove first-order nuisance bias. Orthogonality handles nuisance error; cross-fitting prevents reusing the same noise twice.

6.4.5 Comparison and summary

Motivation. Contrast the R-learner with the DR-learner.

	DR-learner	R-learner
Second stage	regress $\hat{\phi}_i^{DR}$ on Z_i	minimize a residualised weighted loss
Orthogonality	via the EIF / DR score	via Robinson residualisation
Main nuisances	μ_0, μ_1, e	m, e
Model selection	validation MSE for $\hat{\phi}^{DR}$	validation R-loss

DR pseudo-outcome:

$$\hat{\phi}_i^{DR} = \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{X_i\{Y_i - \hat{\mu}_1(Z_i)\}}{\hat{e}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{\mu}_0(Z_i)\}}{1 - \hat{e}(Z_i)}.$$

Validation losses on a held-out set V: DR, $\sum_{i \in V}\{\hat{\phi}_i^{DR} - \hat{\tau}(Z_i)\}^2$; R, $\sum_{i \in V}\{Y_i^r - X_i^r\hat{\tau}(Z_i)\}^2$.

What the R-learner is for. Residualised CATE estimation (estimate $\tau(Z)$ after removing prognosis $m(z)$ and assignment $e(z)$); model selection (tune and compare candidate τ models by validation R-loss); regularized heterogeneity (put shrinkage or structure directly on τ when the effect signal is weak).

Summary.

- R-learning starts from Robinson's decomposition $Y - m(Z) = \{X - e(Z)\}\tau(Z) + \varepsilon$.
- Estimate m and e by cross-fitting, then minimize the residualised R-loss.
- The R-loss is orthogonal to nuisance error and can be optimized with many ML tools.
- The quasi-oracle result says the difficulty lies mainly in the complexity of τ, not of m and e.